

Full length article

Compact polarization beam splitter based on metal-capped thin-film LiNbO₃-on-insulator plasmonic waveguide

Xin-Yang Wang¹, Jia-Min Liu¹, Yu-Jing Yang¹, De-Long Zhang^{*}

School of Precision Instruments and Opto-electronics Engineering, Key Lab of Optoelectronic Information Technology (Ministry of Education), and Key Lab of Micro-Opto-Electro-Mechanical Systems (MOEMS) Technology (Ministry of Education), Tianjin University, Tianjin 300072, China

ARTICLE INFO

Keywords:

Lithium niobate on insulator
Polarization beam splitter
Integrated photonics

ABSTRACT

A compact polarization beam splitter based on a triple-waveguide plasmonic directional coupler is proposed and modeled. The directional coupler consists of two usual lithium-niobate-on-insulator (LNOI) ridge waveguides and a metal-capped plasmonic LNOI waveguide located between them. It utilizes stronger polarization dependence of modal birefringence of a plasmonic waveguide. By appropriately designing the plasmonic waveguide geometry, the TM component wave meets the resonant coupling condition and outputs at cross port. Instead, the TE component wave does not meet the resonant coupling condition, cannot be resonant in the triple-waveguide directional coupler and outputs at through port directly. Excellent performance is expected for an optimal device. First, the device shows an ultra-high polarization extinction ratio (PER) of 28.3 dB and an ultra-low insertion loss (IL) of 0.06 dB for the TE-polarized wave, and of 31.7 dB and of 0.31 dB in order for the TM-polarized wave. Second, the working bandwidth with PER > 25 dB and IL < 0.08 dB is 100 nm (1500–1600 nm) for the TE-polarized wave, and that with PER > 20 dB and IL < 0.4 dB is 30 nm (1534–1564 nm) for the TM-polarized wave. Third, the device has a compact structure of 33 μm in length and exhibits definite fabrication tolerance. Overall, the proposed device exhibits ultra-high PER and ultra-low IL, and has a compact size suitable for on-chip use. A comparison shows that present device exhibits advantage in IL and/or size over that based on other LNOI structures or the SOI platform.

1. Introduction

LiNbO₃ (LN) is one of the most attractive photonic materials because of its excellent electro-optic, acousto-optic and nonlinear properties, as well as the possibility of producing a high-quality waveguide of low-loss. Conventional LN waveguide, fabricated by either titanium in-diffusion or proton exchange technology, has a lower refractive index increment of < 0.02, which results in a looser mode field and a larger bending radius. The use of such LN waveguide leads to a larger footprint of an LN-based photonic device. It is thus not suitable for development of an LN-based photonic integrated circuit (PIC), which monolithically integrates optical components of different functions and is of great merits for uses in many fields, such as ultra-high speed and capacity communication, ultra-sensitive sensing and ultra-fast computing.

With the maturity of crystal ion slicing (smart-cutting) and wafer bonding technologies, a high-quality LiNbO₃-on-insulator (LNOI) thin-film with a larger refractive index contrast of ~ 0.7 and an ultra-low

loss of < 0.03 dB/cm can be prepared at present. Optical waveguide is a building block of a photonic device. A low-loss waveguide can be fabricated on surface of the LNOI thin-film by making use of a technological process combining electron beam lithography (EBL) with reactive ion etching (RIE) and chemical mechanical polishing (CMP) [1,2]. An LNOI waveguide effectively combines the excellent properties of LN crystal itself with the attractive features of LNOI waveguide. The effective combination has stimulated researchers to develop ultra-compact, highly efficient micro-photonic LNOI waveguide devices by utilizing its ultra-small cross-section area (<1 μm²) and bending radius allowed (<10 μm) [2–6]. Over the past years, some highly efficient micro-photonic integrated optical devices based on the LNOI waveguide have been demonstrated. These include electro-optic modulator [7–18], microring [1,2,19–23] and microdisk resonators [24–30], grating coupler [31–33], photonic crystal [34,35], acousto-optic frequency shifter [36], acousto-optic modulator [37], Kerr comb [22,38,39], microring resonator based on periodically poled LNOI waveguide

* Corresponding author.

E-mail address: dlzhang@tju.edu.cn (D.-L. Zhang).

¹ Equally contributed authors.

[40,41], optical metasurface structures [42–44], directional coupler switches based on surface plasmon polaritons (SPPs) [45,46], quantum [6] and polarization management devices [47–55]. Nowadays, the LNOI waveguide has become a promising and attractive material platform for developing various PICs for high-density integrated photonics.

It is well known that an LN-based waveguide device is polarization-dependent because of the intrinsic birefringence property of the LN crystal itself and asymmetry of optical waveguide itself. The electro-optic and nonlinear optical features of LN are also polarization-dependent. As a result, the transverse-electric (TE) and transverse-magnetic (TM) waves propagating along the LN waveguide have different electro-optic and nonlinear effects. This is the case for an LNOI-based waveguide device. One effective approach to solve the problem is polarization diversity. To achieve it, polarization manipulation and management are essential. Polarization beam splitter (PBS) enables to split or combine two orthogonal modes. It is a fundamental building block in a polarization diversity system for polarization manipulation and management, and has been widely used in communication, polarization-division multiplexing and quantum optics. In general, an efficient PBS can be easily realized for a waveguide having a higher modal birefringence, such as the SOI one [56–61]. For the LNOI waveguide, however, it is not the case because of its relatively low modal birefringence. To realize an LNOI PBS, some structural schemes with an IL > 0.3 dB and a device size > 80 μm have been proposed. By using Mach-Zehnder interferometer, stimulated Raman adiabatic passage, *meta*-material and directional coupler, several PBSs based on the LNOI waveguide have been demonstrated in recent years [50–55]. Nevertheless, all of them suffer from this or that problem as described below. 1) Efficient PBSs with ultra-broadband (S + C + L band) and higher polarization extinction ratio (PER) of > 20 dB can be realized by using the adiabatic passage technology and Mach-Zehnder interferometer [50,51]. Nevertheless, their footprints, about thousands of μm^2 , are too large to meet the demand of high-density integration. 2) Hetero-anisotropic *meta*-material enables to achieve a PBS with higher PER (>20 dB) and broader bandwidth of ~ 85 nm. However, the complicated structure increases the difficulty in fabrication [52]. 3) Although the directional coupler (DC) is an ideal scheme, a PBS based on double-waveguide DC has a length > 80 μm [53,54]. Other structural schemes such as sub-wavelength gratings and slot waveguide are also the effective ways to achieve compact footprint and high-performance PBS, and have been demonstrated on the basis of the SOI platform [59–61]. However, it is a challenge to implement a compact PBS with a quite small feature size on the basis of the LNOI platform. In summary, although some PBSs have been demonstrated on the basis of different structural schemes and different material platforms of LNOI and SOI, each suffers from this or that problem [the previously reported LNOI-based PBSs suffer mainly from higher insertion loss (IL) and larger device size, and very strong background absorption of Si material itself causes additional IL of the SOI-based PBSs]. It still needs to explore other ways towards a compact, efficient PBS suitable for on-chip use.

A triple-waveguide plasmonic DC consists of two usual dielectric waveguides and a metal-covered dielectric waveguide located between them [62–65]. Such a DC has a larger modal birefringence. A compact PBS would be realized by making use of it. Thus, a triple-waveguide DC is more promising than the double-waveguide one for realizing a PBS. In this paper, we demonstrate a compact, efficient PBS by using the triple-waveguide plasmonic DC based on the LNOI platform. The DC utilizes stronger polarization dependence of modal birefringence of the plasmonic LNOI waveguide to selectively couple the TM wave on the basis of its resonant coupling condition. The proposed PBS, operated at 1.5 μm , would achieve a PER of 31.7 dB and a insertion loss (IL) of 0.3 dB for the TM component wave, and a PER of 28.3 dB and an IL of 0.06 dB for the TE component wave. Moreover, the proposed PBS device has an operating bandwidth of 100 nm (1500–1600 nm) for the TE wave, over which the PER keeps > 25 dB and the IL retains < 0.08 dB, and of 30 nm (1534–1564 nm) for the TM wave, over which the PER retains > 20 dB

and the IL keeps < 0.4 dB. More important, the proposed device has a short length of 33 μm only suitable for on-chip application in PICs. In addition, the device has definite fabrication tolerance.

2. Device structure and principle

Attention is firstly paid to the structure of the PBS proposed here. Fig. 1(a) shows the three-dimensional (3D) structure of the device. For convenience, an xyz Cartesian coordinate system is fixed at the top surface of the substrate SiO_2 layer and the y axis is parallel to the waveguide axis as indicated in Fig. 1(a). The device is designed on the basis of a Z-cut LNOI platform with a 500-nm-thick LN device layer. It consists of an input port, a through port, a cross port and a triple-waveguide plasmonic DC consisted of coupling and output parts. The coupling part consists of two LNOI ridge waveguides (WG I and WG II), which have the same width w_1 , and a silver-capped plasmonic LNOI ridge waveguide (WG III) with a sub- μm width w_2 . A SiO_2 buffer layer is introduced between the silver and LN layers. Fig. 1(b) shows the cross-section on the xz plane of the coupling part of the triple-waveguide plasmonic DC. The output part of the triple-waveguide plasmonic DC consists of two S-bent LNOI waveguides at the end of WG I and WG II to couple the TE mode to the through port and the TM mode to the cross port.

The relevant geometric parameters are indicated in Fig. 1(a) and 1(b). The input, through, and cross ports have the same width w_1 as that of the WG I and WG II. The parameter h represents the LN layer thickness, which can be controlled in the range of 300–900 nm, depending on the He^+ implantation energy adopted in the fabrication of an LNOI wafer by smart cut (i. e., crystal ion slicing) method [2]. A nearly median thickness value of $h = 500$ nm, which is compatible with that of most LNOI devices, is considered here. The h_A is the thickness of Ag metal cap, and here it is designed as 90 nm as detailed latter. The h_S represents the thickness of the SiO_2 buffer layer. The g represents the gap between the LNOI ridge waveguide WG I or WG II and the Ag-capped LNOI waveguide, and here it is fixed at $g = 200$ nm in order to reduce the fabrication difficulty. The L_C denotes the length of the triple-waveguide DC. The length of the two S-bent LNOI waveguides is $L_S = 9$ μm and their offset in the x direction, named W_S , is fixed at 1.5 μm so as to ensure a minor bending loss. In addition, the sidewall angle (θ) of ridge LNOI waveguide was fixed at $\theta = 80^\circ$ to match with the practical situation of fabrication technology [1]. The refractive indices of LN are $n_o = 2.211$ (ordinary) and $n_e = 2.1376$ (extraordinary) at the 1550 nm wavelength [66], and those of SiO_2 and Ag are 1.445 [67] and $0.1453 + 11.3587i$ [68], respectively.

By utilizing the polarization-dependent nature of plasmonics, a metal-capped plasmonic LNOI sub- μm strip (WG III) is introduced to tailor the polarization dependence of mode refractive index. With appropriately designed geometry of the sub- μm -wide WG III strip, the TM_{00} mode meets the triple-waveguide resonant coupling condition, which is mode phase matching condition in nature as detailed later, while the TE_{00} mode does not meet it and has a considerable phase

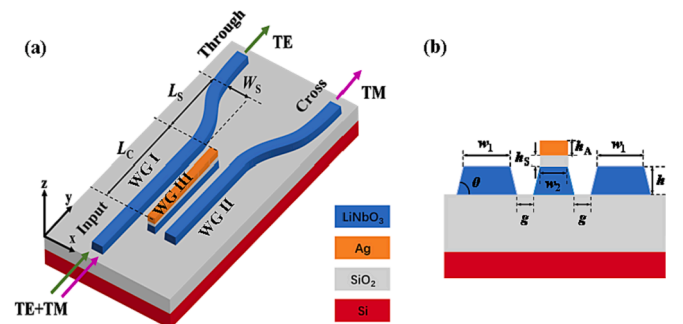


Fig. 1. (a) 3D schematic of proposed PBS (b) cross-section of coupling part.

mismatch. Speaking specifically, the TM_{00} wave efficiently couples among the three waveguides WG I, WGII and WG III, and the resonant coupling length L_C is ultra-short because of quite strong evanescent field coupling. In case that the DC length is designed as the L_C , the TM_{00} wave is resonantly coupled from the input port to the cross port. Because the TE_{00} wave does not meet the resonant condition or phase matching condition, it couples only within the waveguides WG I and WG II. As the separation of the two waveguides WG I and WG II is large enough, $2g + w_2$ as shown in Fig. 1(b), their coupling of evanescent field is relatively weak. The weak coupling results in a coupling length L_C much larger than that in the case of TM_{00} wave. By making use of the quite different resonant coupling lengths of the TE_{00} and TM_{00} waves, one can design an efficient PBS. In case that the coupling length is designed as the ultra-short L_C of the TM_{00} wave, the coupling of TE_{00} wave is so weak that can be ignored. As a result, the TE_{00} wave almost completely passes through the WG I waveguide and fully outputs at the through port. Although the introduction of the WG III strip makes the fabrication process complicated, it provides a new degree of freedom to adjust the polarization state of guided wave, and hence overcomes the shortcoming of lower mode birefringence of the LNOI waveguide.

The wave coupling mechanism of the proposed PBS may be further explained on the basis of the supermodes guided in the coupling region. Fig. 2 shows the electric field profiles of dominant components of all supermodes possibly excited in the triple-waveguide plasmonic DC (at the 1550 nm wavelength). The simulation was carried out in the case that $g = 200$ nm, $w_1 = 700$ nm, $w_2 = 270$ nm, $h_S = 50$ nm, and $h_A = 90$ nm.

In case of TE wave, only one even (symmetric) supermode [see Fig. 2 (a)] and one odd (anti-symmetric) supermode [Fig. 2(b)], which have refractive indices n_{TE1} and n_{TE2} , respectively, are possibly supported by the triple-waveguide DC. In this case the device is equivalent to a usual two-waveguide DC. The TE_{00} wave is resonantly coupled between two waveguides WG I and WG II. The relevant coupling length L_C can be evaluated on the basis of the effective indices of two supermodes using the resonant coupling condition or phase matching condition of the concerned two modes given by coupling mode theory:

$$L_C = \frac{\lambda_0}{2(n_{TE1} - n_{TE2})} = \frac{\lambda_0}{2\Delta n_{TE}} \quad (1)$$

where λ_0 is the working wavelength, and $\Delta n_{TE} = n_{TE1} - n_{TE2}$.

Instead, the TM wave is resonantly coupled among all of the three waveguides. The DC may support three supermodes, including two even supermodes [Fig. 2(c) and 2(d)] and one odd supermode [Fig. 2(e)], which have effective refractive indices of n_{TM1} , n_{TM2} , and n_{TM3} ,

respectively. By designing appropriate device parameters to meet the resonant coupling or mode phase matching condition [65]

$$n_{TM1} + n_{TM2} = 2n_{TM3} \text{ or } n_{TM1} - n_{TM3} = n_{TM3} - n_{TM2} \quad (2)$$

the TM wave can be fully coupled from the input port to the cross port. The relevant coupling length L_C in this case can be estimated by [65]:

$$L_C = \frac{\lambda_0}{2(n_{TM1} - n_{TM3})} = \frac{\lambda_0}{2(n_{TM3} - n_{TM2})} = \frac{\lambda_0}{2\Delta n_{TM}} \quad (3)$$

where $\Delta n_{TM} = n_{TM1} - n_{TM3} = n_{TM3} - n_{TM2}$.

3. Results and discussion

3.1. Design of geometric parameters

To implement efficient coupling of the TM wave meanwhile suppress the coupling of the TE wave and hence realize an efficient PBS device, it needs to design appropriate geometric parameters of the device. First of all, the width w_1 of the two LNOI ridge waveguides is designed. By using Finite-Difference Eigenmode (FDE) solver (Ansys Lumerical MODE Solutions), we have calculated the effective refractive indices of the fundamental and the first order TE and TM modes at 1550 nm wavelength as a function of LNOI ridge waveguide width w_1 . The results are shown in Fig. 3(a). As expected, the mode index increases with the increased width w_1 . The TM_{00} (TE_{00}) mode is cutoff as the waveguide width w_1 is < 290 (360) nm. Moreover, the waveguide supports single-mode propagation in the case of $290 \text{ nm} < w_1 < 1050$ nm for the TM wave and $360 \text{ nm} < w_1 < 940$ nm for the TE wave. An overall consideration for the single-mode operation and the necessity of strong evanescent field of TM_{00} mode for more efficient resonant coupling suggests an LNOI ridge waveguide width $w_1 = 700$ nm.

Attention is next paid to the design of the geometry of the Ag-capped plasmonic LNOI ridge waveguide (WG III), including the width w_2 , SiO_2 buffer layer thicknesses h_S and Ag cap thickness h_A . The w_2 design is on the basis of working principle of the PBS device: the Ag-capped plasmonic LNOI ridge waveguide supports the TM_{00} mode while does not the TE_{00} mode, as well as resonant coupling condition Eq. (2) for the TM wave. First of all, we have calculated the w_2 dependences of effective refractive indices (at 1550 nm wavelength) of TE_{00} and TM_{00} modes guided in the Ag-capped plasmonic LNOI ridge waveguide for different SiO_2 buffer layer thicknesses of $h_S = 30, 40, 50$ and 60 nm. The results are shown in Fig. 3(b). We note that the plasmonic waveguide suppresses the TE mode and supports only the TM_{00} mode as its width w_2 is smaller than 360 nm. We also note that the SiO_2 buffer layer thickness h_S

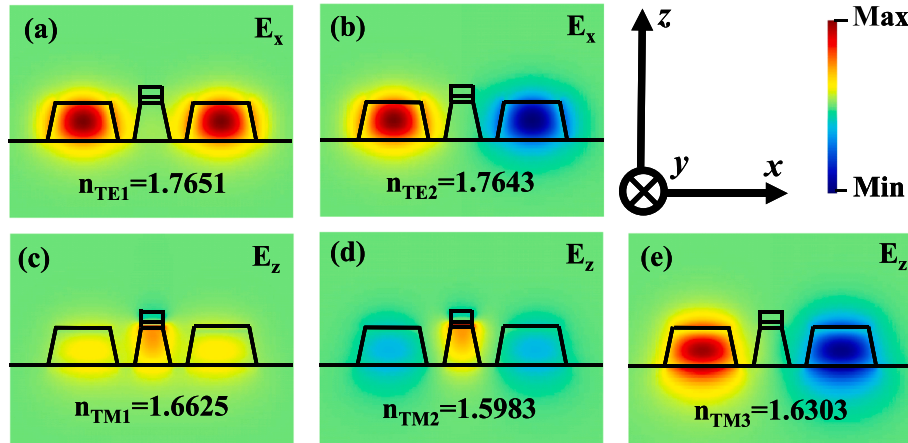


Fig. 2. Electric field profile of dominant components of TE_{00} (E_x) and TM_{00} (E_z) supermodes in a triple-waveguide coupler at 1550 nm wavelength in case that $g = 200$ nm, $w_1 = 700$ nm, $w_2 = 270$ nm, $h_S = 50$ nm and $h_A = 90$ nm. (a) and (b): even and odd TE supermodes, respectively; (c), (d) and (e): even, even and odd TM supermodes in order.

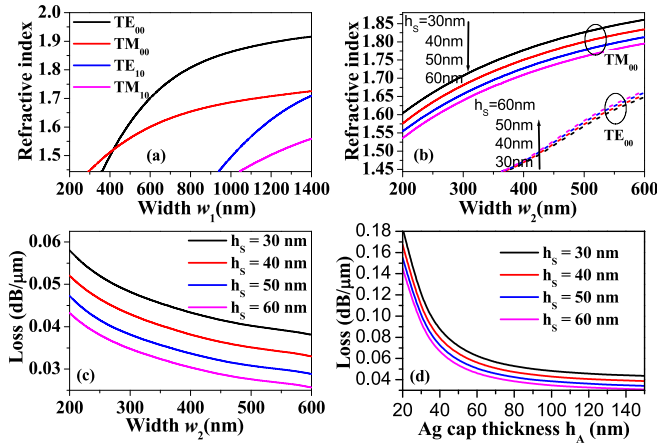


Fig. 3. (a) Effective refractive indices of TE_{00} , TM_{00} , TE_{10} and TM_{10} modes as a function of the LNOI waveguide width w_1 . (b) Effective refractive indices of TE_{00} and TM_{00} modes as a function of Ag-capped plasmonic LNOI waveguide width w_2 for different SiO_2 buffer layer thicknesses of $h_s = 30$ (black), 40 (red), 50 (blue) and 60 nm (magenta). (c) TM_{00} mode loss as a function of Ag-capped plasmonic LNOI waveguide width w_2 in the range of 200–600 nm for different SiO_2 buffer layer thicknesses of $h_s = 30, 40, 50$ and 60 nm. (d) TM_{00} mode loss as a function of Ag cap thickness h_A in the range of 20–150 nm for different SiO_2 buffer layer thicknesses of $h_s = 30, 40, 50$ and 60 nm.

affects slightly the waveguiding characteristic of the TM_{00} mode, whilst hardly that of TE_{00} mode and hence hardly the cut-off condition. These observations suggest a w_2 value of ≤ 360 nm. On the other hand, the w_2 cannot be too small because the waveguide loss increases considerably as the w_2 is decreased as shown in Fig. 3(c), where the propagation loss of the TM_{00} mode supported by the WG III waveguide is plotted against the w_2 for four different SiO_2 buffer layer thicknesses of $h_s = 30, 40, 50$ and 60 nm. In addition, too small w_2 may also cause direct coupling of the TE wave between the two waveguides of WG I and WG II. Appropriate w_2 value is further determined on the basis of the resonant coupling condition given by Eq. (2). Fig. 4 shows the effective indices calculated versus the width w_2 in the range of 200–300 nm for the five supermodes possibly guided in triple-waveguide LNOI DC with the structural parameters of $w_1 = 700$ nm, $h_s = 50$ nm, $h_A = 90$ nm and $g = 200$ nm. For the TM wave, the resonant coupling happens at $w_2 = 270$ nm because the purple $[(n_{TM1} + n_{TM2})/2]$ and blue (n_{TE3}) plots intersect there, which means that Eq. (2) holds as $w_2 = 270$ nm. Instead, the resonant coupling does not happen for the TE wave as the two plots of

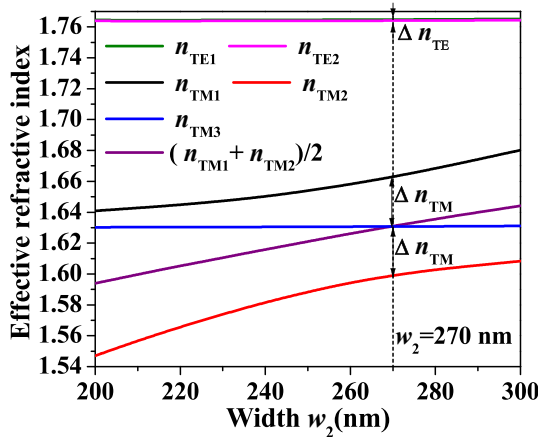


Fig. 4. Ag-capped plasmonic LNOI waveguide width w_2 dependences of effective indices of five supermodes possibly guided in the triple-waveguide LNOI DC with the structural parameters of $w_1 = 700$ nm, $h_s = 50$ nm, $h_A = 90$ nm and $g = 200$ nm.

n_{TE1} and n_{TE2} do not intersect in the w_2 range studied. Thus, here we choose the w_2 as 270 nm, which ensures that the WG III waveguide supports TM_{00} single-mode operation, has a lower propagation loss, and effectively separates WG I and WG II and hence avoids direct cross coupling of the TE wave between these two waveguides.

In addition, in Fig. 4 we indicate the Δn_{TE} and Δn_{TM} at $w_2 = 270$ nm. We can see that $\Delta n_{TE} \ll \Delta n_{TM}$, which leads to a significant difference of coupling length L_C in the two cases of TE and TM waves according to Eq. (1) and Eq. (3). Substituting $\Delta n_{TM} \sim 0.032$ and $\Delta n_{TE} \sim 0.0008$ into the Eq. (1) and Eq. (3), one can find the coupling length $L_C \approx 23.4$ and 970 μm in the two cases of TM and TE waves, respectively. One can see that the L_C of the TM wave is ~ 40 times shorter than that of the TE wave, implying that the coupling of TE wave is weak and can be neglected provided that the coupling length is designed as that of the TM wave. In words, the significantly different supermode distributions of the TE and TM waves in the triple-waveguide DC result in significantly different coupling lengths under the two states of polarization, and the different coupling lengths make it possible to achieve a PBS.

The design of both SiO_2 buffer layer thickness h_s and Ag cap thickness h_A is based on their effects on the propagation loss. As we know, a metal cover layer results in considerable loss of a wave guided in an optical waveguide, and the situation is particularly serious for a TM wave, which is involved in present study. To reduce the loss figure, a buffer layer is usually introduced. Based on the buffer layer thickness dependence of the waveguide loss, one may design an adequate buffer layer thickness. One can see from Fig. 3(c) that the propagation loss of the TM_{00} mode, as expected, decreases considerably with the increased buffer layer thickness h_s for a given width w_2 . On the other hand, as shown in Fig. 3(b), too thick buffer layer results in reduction of the modal birefringence, which is undesired for achieving an efficient PBS device. Based on these considerations, here we choose the h_s as 50 nm. The Ag cap thickness h_A is designed in a similar way. Fig. 3(d) shows the TM_{00} mode loss calculated as a function of the Ag cap thickness h_A in the range of 20–150 nm for different SiO_2 buffer layer thicknesses of $h_s = 30, 40, 50$ and 60 nm. As the Ag cap thickness h_A is increased, the TM_{00} mode loss decreases quickly in the initial stage, slowly then, and tends gradually to a constant for $h_A > 100$ nm. An Ag cap thickness > 80 nm appears to be expected. By taking into account that a thinner metal film is coated easier, here we select an Ag cap thickness of 90 nm, at which the loss in the Ag cap is similar to 0.04 dB/ μm .

The feature that the mode loss decreases with the increased metal thickness may be explained on the basis of the mode field profile feature and the concept of hybrid SPP mode. Fig. 5 shows electric field profile along with depth direction z of the Ag-capped plasmonic waveguide WG

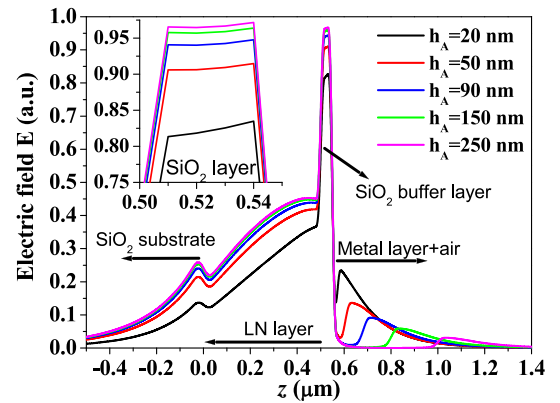


Fig. 5. Electric field profile of waveguide and SPP hybrid mode along with depth direction z of the Ag-capped plasmonic waveguide WG III of the triple-waveguide LNOI DC with structural parameters of $w_1 = 700$ nm, $h_s = 50$ nm, $w_2 = 270$ nm, $g = 200$ nm and varied h_A ranging from 20 to 150 nm. The dielectric or metal medium regions involved are indicated for convenience. Inset shows an expanded view of the mode field in the SiO_2 buffer layer.

III of the triple-waveguide LNOI DC with the structural parameters of $w_1 = 700$ nm, $h_s = 50$ nm, $w_2 = 270$ nm, $g = 200$ nm and $h_A = 20$ – 150 nm. The origin of the reference frame is located at the interface of the LN layer and the SiO_2 substrate. For convenience, the dielectric and metal medium regions involved are indicated that include the SiO_2 substrate, LN, SiO_2 buffer, Ag metal and air medium layers. Inset shows an expanded view of the mode field profile in the SiO_2 buffer layer. We first note that the mode field profile reveals a typical feature of waveguide and SPP hybrid mode. We also note that, in the dielectric layers of LN and SiO_2 buffer medium, the mode field portion increases with the increased Ag thickness. On the other hand, in the Ag layer and air medium covered on its top, the electric field portion of the SPP wave decreases correspondingly with a rise in Ag thickness. These features indicate that, as the metal thickness is increased, the SPP effect weakens and so does its interaction with electromagnetic wave, and hence the mode loss reduces. Here, it is worthwhile to point out that present feature of mode loss versus metal thickness is different from that of the long-range SPP wave, which is excited in a dielectric-metal-dielectric structure with a thin metal layer due to the coupling of two SPP modes excited independently at the two metal/dielectric medium interfaces [69]. The loss of the long-range SPP wave decreases as the metal thickness is decreased, which is distinct from presently observed feature. According to the coupling theory of two SPP modes excited independently at two metal/dielectric medium interfaces of a dielectric-metal-dielectric structure [69], in addition to the long-range SPP mode, another new SPP mode is simultaneously generated due to the coupling, and it reveals a metal thickness dependence of loss similar to the presently observed, i. e., as h_A is increased, the loss decreases strongly for smaller h_A values, slowly then, and gradually tends to a constant.

Finally, the two symmetric S-bent LNOI waveguides are designed. The design is based on the minimal radiation loss, device size and coupling between the two S-bent waveguides. The design gives $L_s = 9$ μm in length and $W_s = 1.5$ μm in lateral offset.

In summary, the geometric parameters are preliminarily designed as $w_1 = 700$ nm, $w_2 = 270$ nm, $h_s = 50$ nm, $h_A = 90$ nm, $g = 200$ nm, $L_c = 23.4$ μm , $L_s = 9$ μm , and $W_s = 1.5$ μm . Among them, there is still an argument to be clarified for the coupling length L_c . The occurrence of coupling between the two S-bent waveguides undoubtedly affects the design of the coupling length L_c , which can be accurately evaluated using Eq. (3) in case without the S-bent waveguides included. To make the argument clear and design an accurate coupling length L_c , we have calculated the coupling efficiency of the TM_{00} mode in case that the coupling length L_c changes in the range of 21–26 μm . In the calculation, the other geometric parameters remain those given above. The calculated results are shown in Fig. 6. We can see that the coupling efficiency

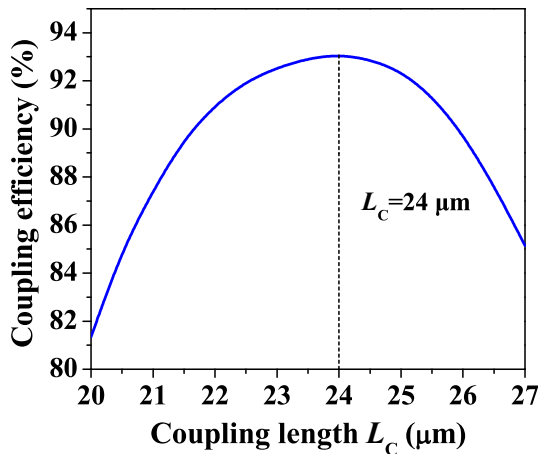


Fig. 6. TM_{00} mode coupling efficiency calculated versus coupling length L_c . The other parameters are $w_1 = 700$ nm, $w_2 = 270$ nm, $h_s = 50$ nm, $h_A = 90$ nm, $g = 200$ nm, $L_s = 9$ μm , and $W_s = 1.5$ μm .

changes considerably with the coupling length L_c and has a maximum value ($\sim 93\%$) at $L_c = 24$ μm . Such a coupling length L_c is similar to that of without the S-bent waveguides included, ~ 23.4 μm , and the coupling efficiency is also similar to 93% at $L_c = 23.4$ μm . This indicates that the coupling between the S-bent waveguides is indeed weak and affects little the coupling efficiency of TM_{00} mode. Moreover, the choice of 270 nm for w_2 effectively suppresses direct coupling between the WG I and WG II waveguides. Therefore, here we choose $L_c = 24$ μm in order to realize a more efficient coupling of the TM_{00} mode.

In conclusion, the geometric parameters are finally designed as $w_1 = 700$ nm, $w_2 = 270$ nm, $h_s = 50$ nm, $h_A = 90$ nm, $g = 200$ nm, $L_c = 24$ μm , $L_s = 9$ μm , and $W_s = 1.5$ μm .

3.2. Performance of device

It is crucial to further evaluate the performance of the PBS device on the basis of the designed optimal geometric parameters given above. For a PBS device, two crucial performance specifications include the PER and IL, which are commonly defined as [70]

$$\text{PER} = 10 \log_{10} \left(\frac{T_{\text{through}}}{T_{\text{cross}}} \right) \text{ for TE component wave,} \quad (4)$$

$$\text{PER} = 10 \log_{10} \left(\frac{T_{\text{cross}}}{T_{\text{through}}} \right) \text{ for TM component wave,} \quad (5)$$

$$\text{IL} = -10 \log_{10} (T_{\text{through}}) \text{ for TE component wave,} \quad (6)$$

$$\text{IL} = -10 \log_{10} (T_{\text{cross}}) \text{ for TM component wave,} \quad (7)$$

in which T_{through} is the transmittance at the through port and T_{cross} is that at the cross port. The performance specification of the PBS device should be defined as smaller one among the PER values of the TE and TM waves while larger one among the IL values of the two polarized waves.

Based on the designed device geometric parameters given above, we have evaluated the two performance specifications PER and IL of the PBS device. Fig. 7 shows the PER and IL calculated as a function of the working wavelength in the range of 1500–1600 nm for the TE_{00} mode (a) and TM_{00} mode (b) guided in the PBS with the optimal geometric parameters given above. For the TE_{00} component wave, as the wavelength is increased from 1500 to 1600 nm, its PER reduces almost linearly from 31.4 dB to 25.3 dB, while its IL increases almost linearly from 0.054 to 0.075 dB as shown in Fig. 7(a). These PER and IL data imply that an ultra-high PER and an ultra-low IL are expected for the TE_{00} component. For the TM_{00} component, as shown in Fig. 7(b), the PER shows a non-monotonous dependence on the wavelength. It has a value > 9.8 dB in the wavelength range considered and peaks at the 1550 nm wavelength. The IL also reveals a non-monotonous wavelength dependence. It changes within the range of 0.31–0.89 dB, which is about one order of magnitude larger than that of the TE_{00} component, and minimizes at the 1550 nm wavelength. In particular, at the 1550 nm wavelength, the PER is 28.3 dB and the IL is 0.06 dB for the TE_{00} component, and 31.7 dB and 0.31 dB in order for the TM_{00} component. In addition, as shown in Fig. 7, the device displays a bandwidth (with $\text{PER} > 25$ dB and $\text{IL} < 0.08$ dB) of 100 nm (1500–1600 nm) for the TE_{00} component wave, and a bandwidth (with $\text{PER} > 20$ dB and $\text{IL} < 0.4$ dB) of 30 nm (1534–1564 nm) for the TM_{00} component wave.

In words, simulation results given in Fig. 7 allow to conclude that the TM_{00} wave is continuously coupled into the cross port via the Ag-capped plasmonic waveguide, and the TM wave energy residual in the input waveguide is minor. Instead, the TE_{00} wave passes almost completely through the LNOI waveguide and outputs fully at the through port. Fig. 8 shows the transmission profile of the electric field $|E|$ (on the xy plane) of the TE_{00} (a) and TM_{00} (b) waves. As shown in Fig. 8(a), the incident TE_{00} component wave directly propagates to the through-port. Instead, the incident TM_{00} component wave outputs at the cross port after an

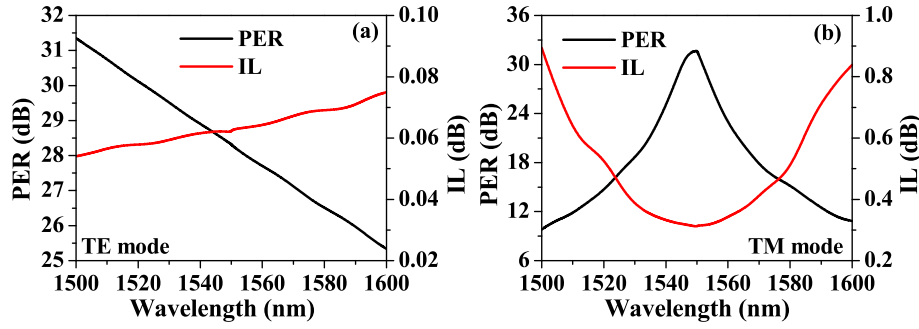


Fig. 7. PER and IL calculated versus wavelength in the range of 1500–1600 nm for (a) TE₀₀ and (b) TM₀₀ mode in the PBS with the optimal geometries of $w_1 = 700$ nm, $w_2 = 270$ nm, $h_S = 50$ nm, $h_A = 90$ nm, $g = 200$ nm, $L_C = 24$ μ m, $L_S = 9$ μ m, and $W_S = 1.5$ μ m.

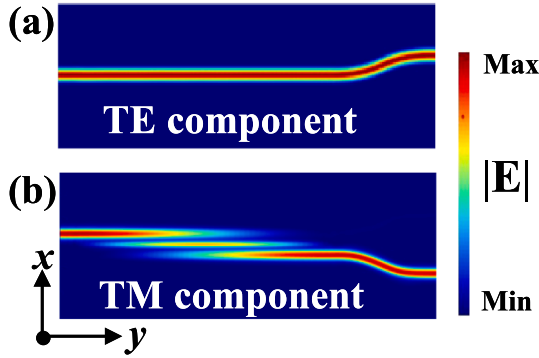


Fig. 8. Transmission profile of electric field $|E|$ at 1550 nm wavelength for (a) TE₀₀ and (b) TM₀₀ modes in the proposed PBS with the optimal geometries of $w_1 = 700$ nm, $w_2 = 270$ nm, $h_S = 50$ nm, $h_A = 90$ nm, $g = 200$ nm, $L_C = 24$ μ m, $L_S = 9$ μ m, and $W_S = 1.5$ μ m.

efficient resonant coupling in the triple-waveguide DC. The transmission profile indicates that the incident TM₀₀ and TE₀₀ component waves can be effectively split by the triple-waveguide DC after a coupling length of $L_C = 24$ μ m. Accordingly, the device length is only 33 μ m in total, including $L_C = 24$ μ m and $L_S = 9$ μ m. As a result, an efficient PBS device is designed.

Here, there is an issue with regard to the Ag cap strip loss to be clarified. As given above, the IL of the device designed here is similar to 0.31 dB in total for the TM₀₀ component wave. It includes the contribution of the metal cap strip, which is about ~ 0.04 dB/ μ m as given above. One may query that the total loss over the entire Ag cap strip length, i. e., the coupling length $L_C = 24$ μ m, is similar to 0.96 dB, which is three times larger than the total device IL value of 0.31 dB. The inconsistency may be explained in terms of an effective propagation distance of the wave in the WG III strip. In the WG III waveguide, not all wave coupled from the WG I waveguide can propagate over the entire coupling length $L_C = 24$ μ m and eventually couples into the WG II waveguide. Part wave couples into the WG II waveguide after a propagation distance in the WG III waveguide shorter than the coupling length $L_C = 24$ μ m. This point is supported by the transmission profile feature of the electric field $|E|$ (on the xy plane) of the TM₀₀ mode guided in the WG III waveguide shown in Fig. 8(b). There exists an effective propagation distance, defined as the full width at half maximum of the profile of the wave intensity along the waveguide axis y . An evaluation gives an effective propagation length of ~ 10 μ m, over which the total loss is ~ 0.4 dB in the WG III strip, comparable to the device IL of 0.31 dB. This indicates that the loss in the Ag cap strip plays a predominant role in the IL of device.

3.3. Tolerance of fabrication

For a device, its geometric parameter values inevitably deviate from those designed ones due to an imperfect fabrication process. It is necessary to carry out a study on fabrication tolerance of the PBS device designed. We have done the study and evaluated the fabrication error tolerance of the proposed PBS. The fabrication error tolerance involves the WG I-III waveguide widths w_1 and w_2 , waveguide gap g , SiO₂ buffer layer thickness h_S , Ag thickness h_A and y -axial position of the WG III waveguide, which have the deviations of Δw_i ($i = 1, 2$), Δg , Δh_S , Δh_A and Δy , respectively. It is reasonably assumed that the WG I-III waveguide widths have the same fabrication error $\Delta w_1 = \Delta w_2 = \Delta w$ that equals the negative value of the waveguide gap deviation, i. e., $\Delta w = -\Delta g$. Fig. 9(a) and 9(b) shows the calculated wavelength dependences of the PER and IL of the TE and TM component waves of the PBS having a waveguide gap deviation of $\Delta g = \pm 5$ nm. Fig. 9(c) and 9(d) shows the case that the SiO₂ buffer layer thickness h_S has a deviation of $\Delta h_S = \pm 5$ nm. Fig. 10(a) and 10(b) shows the case that the Ag cap thickness h_A has a deviation of $\Delta h_A = \pm 10$ nm. Fig. 10(c) and 10(d) shows the case that the WG III waveguide width w_2 has the deviations of $\Delta w_2 = -5, -10$ nm. All calculations were carried out on the basis of the designed optimal geometry of $w_1 = 700$ nm, $w_2 = 270$ nm, $h_S = 50$ nm, $h_A = 90$ nm, $g = 200$ nm, $L_C = 24$ μ m, $L_S = 9$ μ m, and $W_S = 1.5$ μ m.

For the TE component wave, the fabrication error has little effect on the device performance, and this is the case whether the fabrication error originates from the waveguide gap g or the SiO₂ buffer layer thickness h_S , or the Ag cap thickness h_A , or the WG III waveguide width w_2 as shown in Fig. 9(a), 9(c), 10(a) and 10(c).

For the TM component wave, the fabrication error influences slightly the IL of the device. A waveguide gap deviation $\Delta g = +5/-5$ nm induces a slight IL increase from the range of 0.31–0.90 dB [see Fig. 7(b)] to that of (0.32–1.14)/(0.34–1.44) dB [see Fig. 9(b)]. This is also the case for either a $+10/-10$ nm deviation of Ag cap thickness h_A or a 10 nm deviation of a WG III waveguide width w_2 as shown in Fig. 10(b) and 10(d). The IL increases more obviously from 0.31 to 0.90 dB to (0.37–1.61)/(0.44–1.72) dB [Fig. 9(d)] for a $+5/-5$ nm deviation of SiO₂ buffer layer thickness h_S .

Instead, the fabrication error influences the PER significantly. As shown in Fig. 9(b) and 9(d), a $+5/-5$ nm deviation of waveguide gap or SiO₂ buffer layer thickness induces a remarkable red/blue shift of the PER peak, which is designed to be located at 1550 nm as pointed out above. The red shift is similar to 10 nm for $\Delta g = +5$ nm and 37 nm for $\Delta h_S = +5$ nm, and the blue shift is about 25 nm for $\Delta g = -5$ nm and 40 nm for $\Delta h_S = -5$ nm. In the aspect of PER value, the ± 5 nm deviation of Δg (Δh_S) causes the peaking PER value reduction up to 3 (6) dB. The fabrication errors of both the Ag cap thickness and the WG III strip width w_2 affect the PER similarly as shown in Fig. 10(b) and 10(d). But the effects are relatively weak in comparison with those of the Δg and Δh_S . A $+10/-10$ nm deviation of Ag cap thickness h_A induces a PER peak red/blue shift of 6/9 nm and a slight peaking PER value reduction of < 1.4

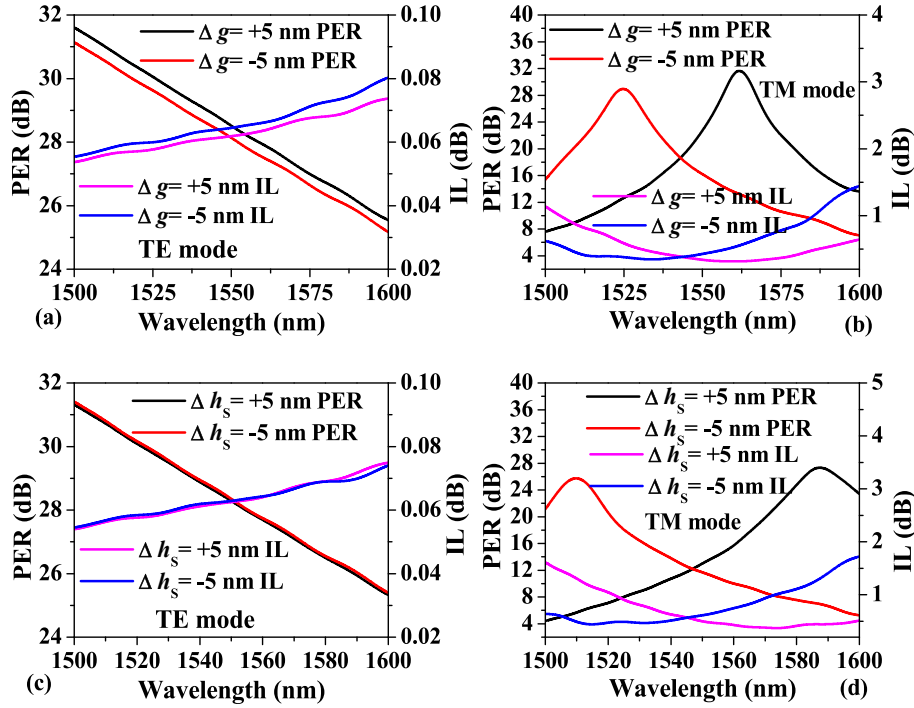


Fig. 9. Calculated PER and IL of the proposed PBS with (a), (b): a gap deviation of $\Delta g = \pm 5$ nm and (c), (d): a SiO₂ buffer layer thickness deviation of $\Delta h_s = \pm 5$ nm. (a), (c): TE mode, (b), (d): TM mode. The calculations aimed at the designed optimal geometry of $w_1 = 700$ nm, $w_2 = 270$ nm, $h_s = 50$ nm, $h_A = 90$ nm, $g = 200$ nm, $L_C = 24$ μ m, $L_S = 9$ μ m, and $W_S = 1.5$ μ m.

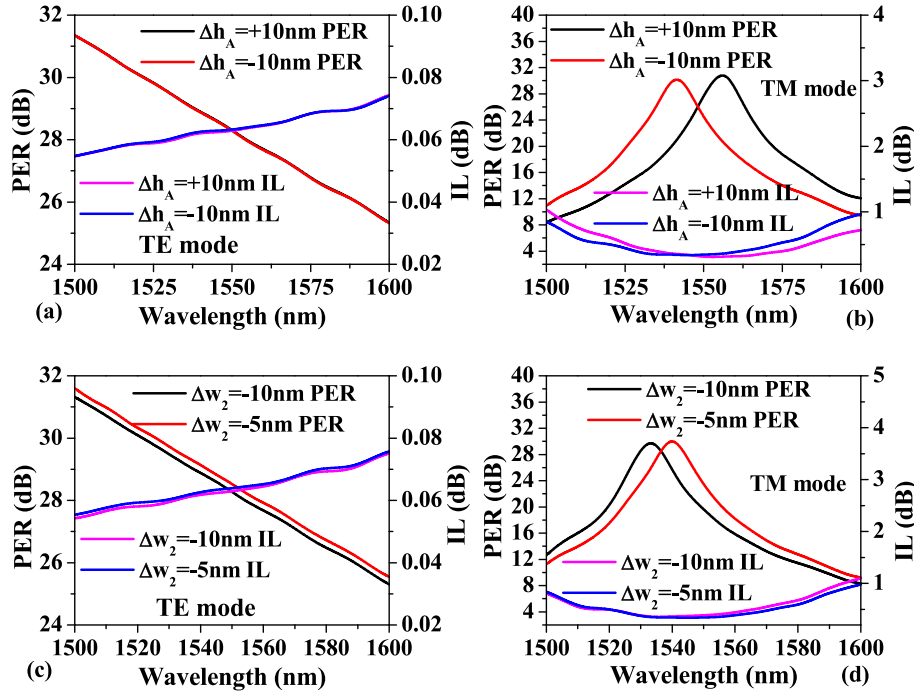


Fig. 10. Calculated PER and IL of the proposed PBS with (a), (b): a Ag cap thickness deviation of $\Delta h_A = \pm 10$ nm and (c), (d): two WG III width deviations of $\Delta w_2 = -5, -10$ nm. (a), (c): TE mode, (b), (d): TM mode. The geometric parameters are the same as those in Fig. 9.

dB. A 10 nm deviation of WG III waveguide width w_2 causes a PER peak shift of 16 nm and a slight peaking PER value reduction of ~ 1.5 dB. Despite of the remarkable shift of the PER peak and definite reduction of peaking PER value, the working bandwidth almost does not change. Additional simulation results show that the central position deviation of $\Delta y = \pm 10$ nm affects little the PBS performance.

Overall, the PBS performance with regard to the TE component wave is little affected by fabrication deviations. For the TM component wave, the fabrication error affects slightly the IL but significantly the PER as indicated by remarkable shift of PER peak and considerable deterioration of peaking PER value. Moreover, the deviation of SiO₂ buffer layer thickness Δh_s affects the PBS performance more obviously than those of

other geometric parameters including the waveguide widths w_1 and w_2 , gap g and Ag cap thickness. Despite of remarkable peak shift and definite reduction of peaking PER value, the working bandwidth is little influenced. The proposed device exhibits definite robustness to the experimental deviation.

3.4. Performance comparison with other PBSs

Finally, it is essential to do a comparison of device performance with other PBSs reported previously to identify the advantages of the PBS proposed in this work. Table 1 brings together the device performance specifications of the PBS proposed here and those of previously reported other ones based on different material platforms of SOI and LNOI and/or on different working principles. These performance specifications include PER, IL, working bandwidth and device length as well. Experimental or simulation work related has been distinguished by the subscript a or b. We firstly do a comparison with the PBSs based on the SOI platform. We can see that present PBS and the SOI one have their respective merits. The SOI PBS has the merits of larger bandwidth and smaller device size. On the other hand, it suffers from larger loss figure. Present device has lower IL, together with the excellent opto-electro, acousto-optic and nonlinear optical properties of LN material. In addition, our device has also sufficiently high PER value and acceptable bandwidth. Note that a PER value similar to 28 dB is already sufficient and an even higher PER value is not absolutely essential. Next, we further do a comparison of our PBS with the ones based on also the LNOI platform but having different structures. One can see that present device has the great merit in the two aspects of IL and device size.

In words, in comparison with the SOI PBSs, present device has a merit in IL, together with the excellent opto-electro, acousto-optic and nonlinear optical properties of LN material itself. Meanwhile, present device has sufficient PER value and acceptable bandwidth. In comparison with other LNOI PBSs, present PBS has the merits in IL and device size.

4. Conclusion

We have demonstrated a compact and efficient PBS based on a triple-waveguide DC consisted of two LNOI ridge waveguides and a Ag-capped LNOI plasmonic waveguide. The introduction of Ag-capped LNOI

plasmonic waveguide presents a new strategy to solve the problem that the LNOI waveguide has a relatively low modal birefringence, which hinders implementation of a compact, efficient PBS. We show that the TM_{00} component wave can fulfill the resonant coupling condition and can be resonantly coupled in the triple-waveguide DC by designing adequate geometric parameters of the Ag-capped LNOI plasmonic waveguide. As a result, the TM_{00} component wave outputs at the cross port after an efficient resonant coupling in the triple-waveguide DC. Instead, the TE_{00} component wave does not meet the resonant condition and hence cannot be resonant in the triple-waveguide DC. Consequently, it directly outputs at the through port. The PBS function is finally achieved.

Optimal geometry of $w_1 = 700$ nm, $w_2 = 270$ nm, $h_s = 50$ nm, $h_A = 90$ nm, $g = 200$ nm, $L_C = 24$ μ m, $L_S = 9$ μ m, and $W_S = 1.5$ μ m is presented for the PBS device proposed. With the optimal geometry, the device displays excellent performance, including an ultra-high PER of 28.3 dB and an ultra-low IL of 0.06 dB for the TE_{00} component wave, and 31.7 dB and 0.3 dB in order for the TM_{00} component wave. The operating bandwidth with a PER > 25 dB and an IL < 0.08 dB is similar to 100 nm (1500–1600 nm) for the TE_{00} component wave, and that with a PER > 20 dB and an IL < 0.4 dB is about 30 nm (1534–1564 nm) for the TM component wave. Moreover, the proposed device has a compact structure of only 33 μ m in length. In words, the proposed PBS device has ultra-high PER, ultra-low IL, compact size suitable for on-chip applications. In addition, the proposed device displays a certain extent of fabrication tolerance. The PBS performance related to the TE component wave is less influenced by the fabrication deviations. For the TM component wave, the fabrication deviation influences slightly the IL but remarkably the PER as indicated by remarkable shift of PER peak and definite reduction of peaking PER value, and the deviation of SiO₂ buffer layer thickness affects the PBS performance more obviously than those of other geometric parameters. Despite of PER peak shift and peaking PER value reduction, the working bandwidth is little influenced by the fabrication errors.

The device IL of the TM_{00} component wave is five times larger than that of the TE_{00} component wave due to the contribution of the loss in the Ag cap strip, which plays a predominant role in the IL of the TM_{00} component wave. The loss in the Ag strip decreases with the increased Ag thickness. The loss feature can be explained in terms of depth profile feature of mode field in the plasmonic waveguide. Simulation shows that

Table 1
A comparison of device performance of various PBSs.

Structure	Material platform	PER (dB)	IL (dB)	Bandwidth (nm)	Length (μ m)	Source
Bent waveguide DC	SOI	~42 (TE) ~30 (TM)	0.8 (TE) 0.3 (TM)	90 (>30 dB, TE) 70 (>30 dB, TM)	26	[57] ^a
MZI	SOI	~25 (TE) ~16 (TM)	–	40 (>10 dB, TM)	~200	[58] ^a
Slot waveguide DC	SOI	~30 (TE) ~40 (TM)	0.17 (TE) 0.22 (TM)	175 (>20 dB, TE) 150 (>25 dB, TM)	11	[59] ^a
Subwavelength gratings DC	SOI	27.5 (TE) 31.9 (TM)	0.12 (TE) 0.14 (TM)	230 (>15 dB, TE) 100 (>20 dB, TM)	6.45	[60] ^a
Plasmonic triple-waveguide DC	SOI	17.6 (TE) 16.5 (TM)	0.54 (TE) 1.29 (TM)	30 (>15 dB, TM)	5.1	[62] ^b
Plasmonic DC	SOI	15.7 (TE) 15.0 (TM)	0.53 (TE) 0.31 (TM)	~120 (>14 dB, TE&TM)	~2.6	[63] ^b
Adiabatic waveguide	x-cut LNOI	~21 (TE) ~45 (TM)	<1.0 (TE&TM)	150 (>20 dB, TE) 250 (>40 dB, TM)	>6000	[51] ^b
MZI	x-cut LNOI	47.7 (TE) 48 (TM)	0.9 (TE) 0.6 (TM)	>200 (>17.8 dB, TE) >200 (>22.8 dB, TM)	~430	[50] ^b
Hetero-anisotropic metamaterial	z-cut LNOI	~24 (TE) ~31 (TM)	0.3 (TE) 1.0 (TM)	185 (>20 dB, TE) 85 (>20 dB, TM)	160	[52] ^b
Mode converter DC	z-cut LNOI	~27 (TE) ~16 (TM)	1.10 (TE) 1.24 (TM)	50 (>26 dB, TE) 50 (>13 dB, TM)	>180	[53] ^a
Plasmonic triple-waveguide DC	z-cut LNOI	28.3 (TE) 31.7 (TM)	0.06 (TE) 0.31 (TM)	100 (>25 dB, TE) 30 (>20 dB, TM)	33	This work ^b

^a Experiment.

^b Simulation.

the mode field depth profile in the plasmonic waveguide reveals a typical feature of waveguide and SPP hybrid mode, and suggests that, as the metal thickness is increased, the SPP effect weakens and so does its interaction with electromagnetic wave, and hence the mode loss decreases. The loss feature can be also associated with coupling of two SPP waves excited independently at two metal interfaces of a dielectric-thin metal-dielectric structure. The coupling generates a new SPP mode that weakens as the metal thickness is increased and reveals a similar metal thickness dependence of mode loss, i. e., the mode loss decreases strongly for smaller metal thickness values, slowly then, and gradually tends to a constant.

In comparison with other LNOI PBSs, present PBS has lower IL and smaller size. In comparison with the SOI PBSs, present device has lower IL, together with the excellent properties of LN material itself. Meanwhile, present device maintains ultra-high PER value and sufficient operating bandwidth.

Present work provides a promising strategy for achieving a compact and efficient PBS based on the LNOI platform for achieving polarization manipulation in LNOI-based PICs. The device may find its uses in fields of on-chip communication, polarization division multiplexing, and polarization entanglement quantum information processing.

CRediT authorship contribution statement

Xin-Yang Wang: Conceptualization, Data curation, Formal analysis, Investigation, Software, Writing – original draft. **Jia-Min Liu:** Data curation, Formal analysis, Investigation, Software, Writing – original draft. **Yu-Jing Yang:** Data curation, Formal analysis, Investigation, Software, Writing – original draft. **De-Long Zhang:** Funding acquisition, Project administration, Resources, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgement

This work was supported by the National Natural Science Foundation of China under Project no. 61875148, and by Key Awards Program of Cultivating Outstanding Innovative Postgraduates in Arts and Sciences of Tianjin University, under Project no. C1-2022-002.

References

- [1] R.B. Wu, M. Wang, J. Xu, J. Qi, W. Fang, Z.W. Fang, J.H. Zhang, J.X. Zhou, L. Qiao, Z.F. Chai, J.T. Lin, Y. Cheng, Long low-loss-lithium niobate on insulator waveguides with sub-nanometer surface roughness, *Nanomaterials* 8 (11) (2018) 910.
- [2] G. Poberaj, H. Hu, W. Sohler, P. Guenter, Lithium niobate on insulator (LNOI) for micro-phonic devices, *Laser Photon. Rev.* 6 (4) (2012) 488–503.
- [3] W. Sohler, R.M. De La Rue, Integrated optics – new material platforms, devices and applications, *Laser Photon. Rev.* 6 (4) (2012) A21–A22.
- [4] A. Boes, B. Corcoran, L. Chang, J. Bowers, A. Mitchell, Status and potential of lithium niobate on insulator (LNOI) for photonic integrated circuits, *Laser Photon. Rev.* 12 (4) (2018) 1700256.
- [5] S. Saravi, T. Pertsch, F. Setzpfandt, Lithium niobate on insulator: an emerging platform for integrated quantum photonics, *Adv. Opt. Mater.* 9 (22) (2021) 2100789.
- [6] Y. Jia, J. Wu, X. Sun, X. Yan, R. Xie, L. Wang, Y. Chen, F. Chen, Integrated photonics based on rare-earth ion-doped thin-film lithium niobate, *Laser Photon. Rev.* 16 (9) (2022) 2200059.
- [7] M.Y. Xu, M.B. He, H.G. Zhang, J. Jian, Y. Pan, X.Y. Liu, L.F. Chen, X.Y. Meng, H. Chen, Z.H. Li, X. Xiao, S.H. Yu, S.Y. Yu, X.L. Cai, High-performance coherent optical modulators based on thin-film lithium niobate platform, *Nat. Commun.* 11 (1) (2020) 3911.
- [8] M.B. He, M.Y. Xu, Y.X. Ren, J. Jian, Z.L. Ruan, Y.S. Xu, S.Q. Gao, S.H. Sun, X. Q. Wen, L.D. Zhou, L. Liu, C.J. Guo, H. Chen, S.Y. Yu, L. Liu, X.L. Cai, High-performance hybrid silicon and lithium niobate Mach-Zehnder modulators for 100 Gbit s⁻¹ and beyond, *Nat. Photonics* 13 (5) (2019) 359–364.
- [9] A. Honardoost, F.A. Juneghani, R. Safian, S. Fathpour, Towards subterahertz bandwidth ultracompact lithium niobate electrooptic modulators, *Opt. Express* 27 (5) (2019) 6495–6501.
- [10] P. Kharel, C. Reimer, K. Luke, L. He, M. Zhang, Breaking voltage-bandwidth limits in integrated lithium niobate modulators using micro-structured electrodes, *Optica* 8 (3) (2021) 357–363.
- [11] A. Rao, S. Fathpour, Compact lithium niobate electrooptic modulators, *IEEE J. Sel. Top. Quant. Electron.* 24 (4) (2018) 3400114.
- [12] C. Wang, M. Zhang, B. Stern, M. Lipson, M. Loncar, Nanophotonic lithium niobate electro-optic modulators, *Opt. Express* 26 (2) (2018) 1547–1555.
- [13] A. Rao, A. Patil, P. Rabiei, A. Honardoost, R. DeSalvo, A. Paoletta, S. Fathpour, High-performance and linear thin-film lithium niobate Mach-Zehnder modulators on silicon up to 50 GHz, *Opt. Lett.* 41 (24) (2016) 5700–5703.
- [14] M. Mahmoud, L.T. Cai, C. Bottenfield, Lithium niobate electro-optic racetrack modulator etched in Y-Cut LNOI platform, *IEEE Photon. J.* 10 (1) (2018) 6600410.
- [15] J.T. Lin, J.X. Zhou, R.B. Wu, M. Wang, Z.W. Fang, W. Chu, J.H. Zhang, L.L. Qiao, Y. Cheng, High-precision propagation-loss measurement of single-mode optical waveguides on lithium niobate on insulator, *Micromachines* 10 (9) (2019) 612.
- [16] M.W. Jin, J.Y. Chen, Y.M. Sua, Y.P. Huang, High-extinction electro-optic modulation on lithium niobate thin film, *Opt. Lett.* 44 (5) (2019) 1265–1268.
- [17] A.J. Mercante, P. Yao, S.Y. Shi, G. Schneider, J. Murakowski, D.W. Prather, 110 GHz CMOS compatible thin film LiNbO₃ modulator on silicon, *Opt. Express* 24 (14) (2016) 15590–15595.
- [18] C. Wang, M. Zhang, X. Chen, M. Bertrand, A. Shams-Ansari, S. Chandrasekhar, P. Winzer, M. Loncar, Integrated lithium niobate electro-optic modulators operating at CMOS-compatible voltages, *Nature* 562 (7725) (2018) 101–104.
- [19] A. Guarino, G. Poberaj, D. Rezzonico, R. Degl'Innocenti, P. Gunter, Electro-optically tunable microring resonators in lithium niobate, *Nat. Photon.* 1 (2007) 407–410.
- [20] R. Luo, Y. He, H.X. Liang, M.X. Li, J.W. Ling, Q. Lin, Optical parametric generation in a lithium niobate microring with modal phase matching, *Phys. Rev. Appl.* 11 (3) (2019), 034026.
- [21] M. Zhang, B. Buscaino, C. Wang, A. Shams-Ansari, C. Reimer, R.R. Zhu, J.M. Kahn, M. Loncar, Broadband electro-optic frequency comb generation in a lithium niobate microring resonator, *Nature* 568 (7752) (2019) 373–377.
- [22] C. Wang, M. Zhang, M.J. Yu, R.R. Zhu, H. Hu, M. Loncar, Monolithic lithium niobate photonic circuits for Kerr frequency comb generation and modulation, *Nature Commun.* 10 (2019) 978.
- [23] B. Desiatov, A. Shams-Ansari, M. Zhang, C. Wang, M. Loncar, Ultra-low-loss integrated visible photonics using thin-film lithium niobate, *Optica* 6 (3) (2019) 380–384.
- [24] S.J. Liu, Y.L. Zheng, X.F. Chen, Cascading second-order nonlinear processes in a lithium niobate-on-insulator microdisk, *Opt. Lett.* 42 (18) (2017) 3626–3629.
- [25] Z.W. Fang, Y.X. Xu, M. Wang, L.L. Qiao, J.T. Lin, W. Fang, Y. Cheng, Monolithic integration of a lithium niobate microresonator with a free-standing waveguide using femtosecond laser assisted ion beam writing, *Sci. Rep.* 7 (2017) 45610.
- [26] L. Wang, C. Wang, J. Wang, F. Bo, M. Zhang, Q. Gong, M. Loncar, Y.F. Xiao, High-Q chaotic lithium niobate microdisk cavity, *Opt. Lett.* 43 (12) (2018) 2917–2920.
- [27] S.J. Liu, Y.L. Zheng, Z.W. Fang, X.N. Ye, Y. Cheng, X.F. Chen, Effective four-wave mixing in the lithium niobate on insulator microdisk by cascading quadratic processes, *Opt. Lett.* 44 (6) (2019) 1456–1459.
- [28] M. Wang, Y. Xu, Z. Fang, Y. Liao, P. Wang, W. Chu, L. Qiao, J. Lin, W. Fang, Y. Cheng, On-chip electro-optic tuning of a lithium niobate microresonator with integrated in-plane microelectrodes, *Opt. Express* 25 (1) (2017) 124–129.
- [29] R.B. Wu, J.H. Zhang, N. Yao, W. Fang, L.L. Qiao, Z.F. Chai, J.T. Lin, Y. Cheng, Lithium niobate micro-disk resonators of quality factors above 10⁷, *Opt. Lett.* 43 (17) (2018) 4116–4119.
- [30] Y.L. Zheng, Z.W. Fang, S.J. Liu, Y. Cheng, X.F. Chen, High-Q exterior whispering-gallery modes in a double-layer crystalline microdisk resonator, *Phys. Rev. Lett.* 122 (25) (2019), 253902.
- [31] S. Kang, R. Zhang, Z. Hao, D. Jia, F. Gao, F. Bo, G. Zhang, J. Xu, High-efficiency chirped grating couplers on lithium niobate on insulator, *Opt. Lett.* 45 (24) (2020) 6651–6654.
- [32] Z.H. Chen, R.H. Peng, Y.W. Wang, H.B. Zhu, H. Hu, Grating coupler on lithium niobate thin film waveguide with a metal bottom reflector, *Opt. Mater. Exp.* 7 (11) (2017) 4010–4017.
- [33] A. Kar, M. Bahadori, S.B. Gong, L.L. Goddard, Realization of alignment-tolerant grating couplers for z-cut thin-film lithium niobate, *Opt. Exp.* 27 (11) (2019) 15856–15867.
- [34] L.T. Cai, H.P. Han, S.M. Zhang, H. Hu, K.M. Wang, Photonic crystal slab fabricated on the platform of lithium niobate-on-insulator, *Opt. Lett.* 39 (7) (2014) 2094–2096.
- [35] H.X. Liang, R. Luo, Y. He, H.W. Jiang, Q. Lin, High-quality lithium niobate photonic crystal nanocavities, *Optica* 4 (10) (2017) 1251–1258.
- [36] L.B. Shao, N. Sinclair, J. Leatham, Y.W. Hu, M.J. Yu, T. Turpin, D. Crowe, M. Loncar, Integrated microwave acousto-optic frequency shifter on thin-film lithium niobate, *Opt. Express* 28 (16) (2020) 23728–23738.
- [37] L.B. Shao, M.J. Yu, S. Maity, N. Sinclair, L. Zheng, C. Chia, A. Shams-Ansari, C. Wang, M. Zhang, K.J. Lai, M. Loncar, Microwave-to-optical conversion using lithium niobate thin-film acoustic resonators, *Optica* 6 (12) (2019) 1498–1505.

- [38] Y. He, Q.F. Yang, J.W. Ling, R. Luo, H.X. Liang, M.X. Li, B.Q. Shen, H.M. Wang, K. Vahala, Q. Lin, Self-starting bi-chromatic LiNbO₃ soliton microcomb, *Optica* 6 (9) (2019) 1138–1144.
- [39] Z. Gong, X.W. Liu, Y.T. Xu, M.R. Xu, J.B. Surya, J.J. Lu, A. Bruch, C.L. Zou, H. X. Tang, Soliton microcomb generation at 2 μ m in z-cut lithium niobate microring resonators, *Opt. Lett.* 44 (12) (2019) 3182–3185.
- [40] J.Y. Chen, Z.H. Ma, Y.M. Sua, Z. Li, C. Tang, Y.P. Huang, Ultra-efficient frequency conversion in quasi-phase-matched lithium niobate microrings, *Optica* 6 (9) (2019) 1244–1245.
- [41] J.J. Lu, M. Li, C. Zou, A.A. Sayem, H. Tang, Towards 1% single photon anharmonicity with periodically-poled lithium niobate microring resonators, *Optica* 7 (12) (2020) 1654–1659.
- [42] B.F. Gao, M.X. Ren, W. Wu, H. Hu, W. Cai, J.J. Xu, Lithium niobate metasurfaces, *Laser Photon. Rev.* 13 (5) (2019) 1800312.
- [43] C. Wang, Z. Li, M.H. Kim, X. Xiong, X.F. Ren, G.C. Guo, N. Yu, M. Loncar, Metasurface-assisted phase-matching-free second harmonic generation in lithium niobate waveguides, *Nat. Commun.* 8 (1) (2017) 2098.
- [44] B. Fang, H.M. Li, S.N. Zhu, T. Li, Second-harmonic generation and manipulation in lithium niobate slab waveguides by grating metasurfaces, *Photonics Res.* 8 (8) (2020) 1296–1300.
- [45] M. Thomaschewski, V.A. Zenin, C. Wolff, S.I. Bozhevolnyi, Plasmonic monolithic lithium niobate directional coupler switches, *Nat. Commun.* 11 (1) (2020) 748.
- [46] S. Slussarenko, G.J. Pryde, Photonic quantum information processing: a concise review, *Appl. Phys. Rev.* 6 (4) (2019), 041303.
- [47] Y. Liu, X. Huang, Z. Li, Y. Kuang, H. Guan, Q. Wei, Z. Fan, Z. Li, TE/TM-pass polarizers based on lateral leakage in a thin film lithium niobate-silicon nitride hybrid platform, *Opt. Lett.* 45 (17) (2020) 4915–4918.
- [48] Z. Chen, J. Yang, W.-H. Wong, E.-Y.-B. Pun, C. Wang, Broadband adiabatic polarization rotator-splitter based on a lithium niobate on insulator platform, *Photon. Res.* 9 (12) (2021) 2319–2324.
- [49] X. Wang, A. Pan, T. Li, C. Zeng, J. Xia, Efficient polarization splitter-rotator on thin-film lithium niobate, *Opt. Express* 29 (23) (2021) 38044–38052.
- [50] H. Xu, D. Dai, L. Liu, Y. Shi, Proposal for an ultra-broadband polarization beam splitter using an anisotropy-engineered Mach-Zehnder interferometer on the x-cut lithium-niobate-on-insulator, *Opt. Express* 28 (8) (2020) 10899–10908.
- [51] G. Yang, A.V. Sergienko, A. Ndao, Plasmonic loss-mitigating broadband adiabatic polarizing beam splitter, *Opt. Lett.* 47 (3) (2022) 629–632.
- [52] C. Deng, M. Lu, Y. Sun, L. Huang, D. Wang, G. Hu, R. Zhang, B. Yun, Y. Cui, Broadband and compact polarization beam splitter in LNOI hetero-anisotropic metamaterials, *Opt. Express* 29 (8) (2021) 11627–11634.
- [53] Y. Wu, X. Sun, H. Li, C. Lu, Y. Zhang, S. Liu, Y. Zheng, X. Chen, Lithium niobate thin film polarization beam splitter based on asymmetric directional coupling, *J. Lightwave Technol.* 42 (24) (2022) 7843–7847.
- [54] Z. Gong, R. Yin, W. Ji, J. Wang, C. Wu, X. Li, S. Zhang, Optimal design of DC-based polarization beam splitter in lithium niobate on insulator, *Opt. Commun.* 396 (2017) 23–27.
- [55] L. Zhang, L. Zhang, X. Fu, L. Yang, Compact, broadband and low-loss polarization beam splitter on lithium-niobate-on-insulator using a silicon nanowire assisted waveguide, *IEEE Photonics J.* 12 (5) (2020) 1–6.
- [56] H. Fukuda, K. Yamada, T. Tsuchizawa, T. Watanabe, H. Shinojima, S.I. Itabashi, Ultrasmall polarization splitter based on silicon wire waveguides, *Opt. Express* 14 (25) (2006) 12401–12408.
- [57] J.R. Ong, T.Y.L. Ang, E. Sahin, B. Pawlina, G.F.R. Chen, D.T.H. Tan, S.T. Lim, C. E. Png, Broadband silicon polarization beam splitter with a high extinction ratio using a triple-bent-waveguide directional coupler, *Opt. Lett.* 42 (21) (2017) 4450–4453.
- [58] D. Dai, Z. Wang, J. Peters, J.E. Bowers, Compact polarization beam splitter using an asymmetrical Mach-Zehnder interferometer based on silicon-on-insulator waveguides, *IEEE Photon. Technol. Lett.* 24 (8) (2012) 673–675.
- [59] Y. Tian, J. Qiu, C. Liu, S. Tian, Z. Huang, J. Wu, Compact polarization beam splitter with a high extinction ratio over S+C+L band, *Opt. Express* 27 (2) (2019) 999–1009.
- [60] S. Dai, J. Xiao, Compact and broadband silicon-based polarization beam splitter using asymmetric directional couplers embedded with subwavelength gratings and slots, *Appl. Opt.* 61 (1) (2022) 126–134.
- [61] J.J. Zhang, J. Yang, L. Liang, W. Wu, Broadband TM-mode-pass polarizer and polarization beam splitter using asymmetrical directional couplers based on silicon subwavelength grating, *Opt. Commun.* 407 (2018) 46–50.
- [62] F. Lou, D. Dai, L. Wosinski, Ultracompact polarization beam splitter based on a dielectric-hybrid plasmonic-dielectric coupler, *Opt. Lett.* 37 (16) (2012) 3372–3374.
- [63] B. Ni, J. Xiao, Ultracompact and broadband silicon-based polarization beam splitter using an asymmetrical directional coupler, *IEEE J. Quantum Electron.* 53 (4) (2017) 8400208.
- [64] Y. Zhao, W. Yu, S. Dai, J. Li, T. Lin, M. Li, J. Liu, Compact lithium-niobate-on-insulator polarization rotator based on asymmetric hybrid plasmonics waveguide, *IEEE Photonics J.* 13 (4) (2021) 1–5.
- [65] J.P. Donnelly, H.A. Haus, N. Whitaker, Symmetric three-guide optical coupler with nonidentical center and outside guides, *IEEE J. Quantum Electron.* 23 (4) (1987) 401–406.
- [66] E. Zelmon, D.L. Small, D. Jundt, Infrared corrected Sellmeier coefficients for congruently grown lithium niobate and 5 mol. % magnesium oxide-doped lithium niobate, *J. Opt. Soc. Am. B* 14 (12) (1997) 3319–3322.
- [67] H. Malitson, Interspecimen comparison of the refractive index of fused silica, *J. Opt. Soc. Am.* 55 (10) (1965) 1205–1208.
- [68] P.B. Johnson, R.W. Christy, Optical constants of the noble metals, *Phys. Rev. B* 6 (12) (1972) 4370–4379.
- [69] D. Sarid, Long-range surface-plasma waves on very thin metal films, *Phys. Rev. Lett.* 47 (26) (1981) 1927–1930.
- [70] H. Xu, D. Dai, Y. Shi, Ultra-broadband and ultra-compact on-chip silicon polarization beam splitter by using hetero-anisotropic metamaterials, *Laser Photon. Rev.* 13 (4) (2019) 1800349.